LESSON 05 · CHEAT SHEET

Strings, Files & CSV

SPLIT & JOIN

```
"Pennyworth, Nye".split(", ")  ['Pennyworth', 'Nye']
"a b  c".split()               ['a', 'b', 'c']
"x;y;z".split(";")             ['x', 'y', 'z']
", ".join(["a", "b"])          'a, b'
text.splitlines()              # list of lines
last, first = s.split(", ")    # unpack
sorted(set(items))             # unique, sorted
```

WRITE A TEXT FILE

```
with open("my_work/out.txt", "w",
          encoding="utf-8") as f:
    f.write("one line\n")      # add \n yourself
    f.write(f"total: {x}\n")
```

`"w"`	write (replaces the file!)
`"a"`	append to the end
`"r"`	read (the default)

READ A TEXT FILE

```
with open("my_work/out.txt",
          encoding="utf-8") as f:
    text = f.read()            # whole file, one str

with open(path, encoding="utf-8") as f:
    for line in f:             # line by line
        line = line.strip()    # drop the \n

lines = f.readlines()          # list of lines
```

The `with` block closes the file for you. Indent everything that needs the file.

READ A CSV

```
import csv
with open("data/x.csv", newline="",
          encoding="utf-8") as f:
    reader = csv.DictReader(f)
    print(reader.fieldnames)   # header
    for row in reader:         # row = dict
        print(row["Salary"])   # always TEXT
    rows = list(reader)        # or keep them all
```

Values arrive as strings: `int()` / `float()` them. Like VBA's `Open … For Input` + `Split`, minus the pain.

WRITE A CSV

```
with open("my_work/clean.csv", "w",
          newline="", encoding="utf-8") as f:
    w = csv.DictWriter(f, fieldnames=["id", "name"])
    w.writeheader()
    w.writerow({"id": "TM1", "name": "Jen"})
    w.writerows(list_of_dicts)
```

CLEANING RECIPES

```
s.strip().upper()          # " tm1 " -> "TM1"
s.replace("cr", "").replace(",", "")
float(s) if s != "" else 0.0
s.replace("hrs", "").strip()
first.strip().title()      # "MAREN" -> "Maren"
d[k] = d.get(k, 0) + v     # count into a dict
```

PATHLIB

```
from pathlib import Path
p = Path("data") / "roll.csv"      # joins with /
p.name        'roll.csv'
p.exists()    True
p.read_text(encoding="utf-8")      # whole file
p.write_text("hi\n", encoding="utf-8")
for p in sorted(Path("data").glob("*.csv")):
    ...                            # every CSV
```

GOTCHAS

- `"w"` wipes the file before writing. Never open `data/` files with `"w"`
- CSV: always `newline=""` and `encoding="utf-8"`
- Don't `.split(",")` a CSV line: "Pennyworth, Nye" has a comma inside the quotes
- Run scripts from the lesson folder, or `data/…` isn't found
- Forgot the `\n`? Everything lands on one line